Uni-mHC: Unistochastic Manifold-Constrained Hyper-Connections with Quantum-Interference Disentanglement

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Abstract

Manifold-constrained hyper-connections (mHC) stabilize deep multi-stream residual training by projecting the inter-stream mixing matrix onto the Birkhoff polytope of doubly stochastic matrices via Sinkhorn–Knopp iterations. Entry-wise non-negativity, however, restricts mixing to *additive convex combinations*: features can never be subtracted, a limitation recently criticized by the spectral-sphere-constrained hyper-connections (sHC). We propose **Uni-mHC**, which lifts the constraint manifold from the Birkhoff polytope to the *unistochastic* set: a complex Cayley transform of a skew-Hermitian parameter matrix yields a unitary U whose modulus square $S = |U| \odot |U|$ is doubly stochastic *by construction*—no projection, no iteration. Macroscopically S preserves exact energy conservation; microscopically a heterodyne readout $W = \mathrm{Re}(Ue^{i\phi})$ admits negative effective weights through destructive (phase-π) interference while remaining row-wise non-expansive. On a controlled subtractive task ($Y = X_0 - X_1$), non-negative baselines plateau at the theoretical floor $\frac{1}{2}\,\mathrm{Var}(Y) \approx 0.99$ whereas Uni-mHC converges to 1.1×10^{-11}; the operator is also the fastest and constraint-exact of all compared parameterizations, and the identity-initialized four-stream topology—instantiated here with Uni-mHC—reaches $2.3\times$ lower validation loss than an identically trained vanilla GPT.

1 Introduction

Residual streams are the backbone of modern deep networks, from the two-path residual connection (He et al., 2016) to hyper-connections (HC) (Zhu et al., 2024), which maintain n parallel streams with a learned mixing matrix at every block boundary. Unconstrained mixing can amplify signal exponentially across depth, so mHC (DeepSeek-AI, 2025) constrains it to the Birkhoff polytope of doubly stochastic matrices (Sinkhorn, 1964; Idel and Wolf, 2015).

The Birkhoff constraint carries an expressivity tax: since $H_{jk} \geq 0$, every stream update is a convex combination—mixing can only *add*, never *subtract*. The sHC critique (Liu et al., 2026) argues that this purely additive coupling pushes deep stacks toward identity degeneration; yet sHC's spectral sphere only bounds worst-case amplification, giving up exact per-row/column energy bookkeeping. We resolve this dichotomy by moving from real to complex manifolds (Fig. 1): a complex Cayley transform gives a unitary U whose $S = |U|^{\odot 2}$ is *exactly* doubly stochastic, while a heterodyne readout $W = \mathrm{Re}(U\,\mathrm{diag}\,e^{i\phi})$ restores subtractive arithmetic through phase interference. Contributions: (i) an exact unistochastic operator, with proofs of doubly stochasticity and unconditional well-conditioning; (ii) an impossibility floor $\frac{1}{2}\,\mathrm{Var}(Y)$ for all entrywise non-negative mixers, and an exact subtraction theorem for ours; (iii) operator benchmarks, the disentanglement experiment, and a four-stream nanoGPT integration (Sec. 3).

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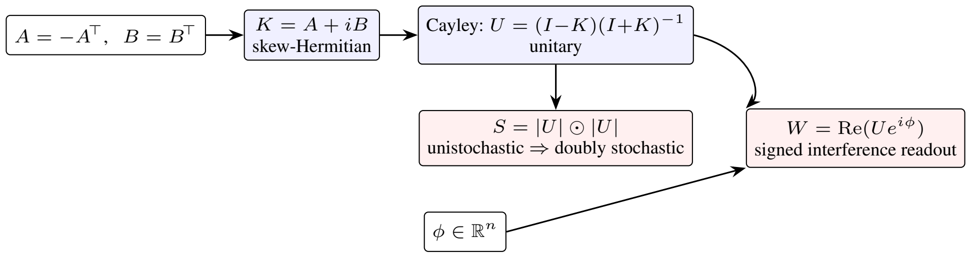

Figure 1: **Uni-mHC operator.** The complex Cayley transform of a skew-Hermitian K is unitary; $S = |U|^{\odot 2}$ is the doubly stochastic energy bookkeeping, while the heterodyne readout W is the signed mixing applied to streams.

2 Mathematical Theory

Setup. A hyper-connected network maintains n residual streams ($n = 4$ here); at every block boundary a mixing matrix $M \in \mathbb{R}^{n\times n}$ redistributes streams. mHC requires the mixer's *energy* bookkeeping matrix to be doubly stochastic so that no stream's energy is amplified or leaked.

Definition 1 (Uni-mHC operator)**.** *Let $\theta_A, \theta_B \in \mathbb{R}^{n\times n}$ be free parameters and $\phi \in \mathbb{R}^n$ per-stream heterodyne phases. Set*

$$A = \theta_A - \theta_A^\top, \qquad B = \theta_B + \theta_B^\top, \qquad K = A + iB, \tag{1}$$

and define the unitary and the two mixing matrices

$$U = (I-K)(I+K)^{-1}, \qquad S = \mathrm{Re}(U)^{\odot 2} + \mathrm{Im}(U)^{\odot 2}, \qquad W = \mathrm{Re}\big(U\,\mathrm{diag}(e^{i\phi})\big). \tag{2}$$

S is the (unistochastic) energy matrix; W is the effective signed mixing applied to the streams.

Theorem 1 (Exact unistochasticity)**.** *For any θ_A, θ_B: (i) K in* (1) *is skew-Hermitian; (ii) $I + K$ is invertible and U in* (2) *is unitary; (iii) $S = |U|^{\odot 2}$ is doubly stochastic; (iv) every singular value of $I + K$ is at least* 1*, so the Cayley solve is unconditionally well-conditioned.*

Proof. (i) $K^H = \overline{A+iB}^{\top} = A^\top - iB^\top = -A - iB = -K$. (ii) K is normal with purely imaginary spectrum $i\lambda_1, \ldots, i\lambda_n$, $\lambda_j \in \mathbb{R}$; hence $I + K$ has eigenvalues $1 + i\lambda_j \neq 0$. Since $(I+K)(I-K) = I - K^2 = (I-K)(I+K)$, the inverses commute, and $UU^H = (I-K)\big[(I+K)^H\big]^{-1}(I-K)^H = (I-K)(I-K)^{-1}(I+K)^{-1}(I+K) = I$, using $(I+K)^H = I - K$; likewise $U^H U = I$. (iii) $\sum_k |U_{jk}|^2 = (UU^H)_{jj} = 1$ and $\sum_j |U_{jk}|^2 = (U^H U)_{kk} = 1$, and $S \geq 0$ entrywise: S is unistochastic, hence doubly stochastic. (iv) Singular values of $I + K$ are $|1 + i\lambda_j| = \sqrt{1+\lambda_j^2} \geq 1$. □

Unlike SK projection, which needs T iterations and leaves a residual column-sum error (measured 5×10^{-3} after $T = 20$, Sec. 3.1), Theorem 1 holds exactly by construction, and the operator costs one $n \times n$ complex linear solve; the Cayley map is a diffeomorphism onto a dense open subset of $\mathrm{U}(n)$, so the parameterization is lossless and smooth, and gradients (through `torch.linalg.solve` in `complex64`) were finite in all experiments.

Proposition 1 (Non-expansive heterodyne mixing)**.** *Writing $U_{jk} = |U_{jk}|e^{i\theta_{jk}}$, we have $W_{jk} = |U_{jk}|\cos(\theta_{jk} + \phi_k)$, so $|W_{jk}| \leq |U_{jk}|$ and $\sum_k W_{jk}^2 \leq \sum_k |U_{jk}|^2 = 1$ for every row, whence $\|Wx\|_2 \leq \|x\|_2$: mixing never amplifies energy, while individual weights become negative whenever a phase shifts by π (destructive interference).*

Theorem 2 (Exact subtractive disentanglement)**.** *There exist parameters of Definition 1 such that row 0 of W equals $\frac{1}{\sqrt{2}}(1, -1, 0, \ldots, 0)$ while S remains doubly stochastic; with readout gain $\sqrt{2}$ the layer computes $X_0 - X_1$ exactly.*

Proof. Take A with a single block $\left(\begin{smallmatrix} 0 & -a \\ a & 0 \end{smallmatrix}\right)$, $a = \tan\frac{\pi}{4}$, $B = 0$; the Cayley transform of a rotation generator is the rotation with $\tan\frac{\theta}{2} = a$, i.e. $\theta = \frac{\pi}{2}$, and row 0 of $U = R(\theta) \oplus I_{n-2}$ is

Table 1: **Operator benchmark** ($n = 4$): worst doubly-stochastic violation of S over 100 random draws, and 1000 forward+backward passes at $(B, T, d) = (32, 256, 384)$.

Operator	row err	col err	params	ms/iter	peak MB
UniMHC (Ours)	4.8e-07	7.2e-07	36	10.56	352
GivensMHC (Ours-Fast)	3.6e-07	3.6e-07	10	10.85	352
Sinkhorn mHC (T=20)	1.2e-07	5.0e-03	16	12.93	352
Orthostochastic (NS)	2.4e-07	2.4e-07	16	11.23	352

$\frac{1}{\sqrt{2}}(1, 1, 0, \dots, 0)$ (sign of a chosen accordingly). Set $\phi = (0, \pi, 0, \dots, 0)$. Then $W_{00} = \mathrm{Re}(U_{00}) = \frac{1}{\sqrt{2}}$ and $W_{01} = \mathrm{Re}(U_{01}e^{i\pi}) = -\frac{1}{\sqrt{2}}$; the other entries of row 0 vanish. Doubly stochasticity of $S = |U|^{\odot 2}$ holds by Theorem 1 for every parameter value, including this one. $\square$

Proposition 2 (Impossibility floor for non-negative mixers). *Let $X_0, \dots, X_{n-1}$ be i.i.d. zero-mean unit-variance and $Y = X_0 - X_1$. For any readout c whose entries share a sign—the cone induced by every entrywise non-negative mixing row, including Sinkhorn/Birkhoff projections and orthostochastic readouts $Q \odot Q$—,*

$$\min_{c\,\text{same sign}} \; \mathbb{E}\left\| \textstyle\sum_k c_k X_k - Y \right\|^2 \;=\; 1 \;=\; \tfrac{1}{2}\,\mathrm{Var}(Y). \tag{3}$$

Proof. With $t = (1, -1, 0, \dots, 0)$, $\mathbb{E}\|\sum_k c_k X_k - Y\|^2 = \|c - t\|_2^2$ by independence and unit variance. The minimizer $c^\star = t$ has mixed signs. On the all-nonnegative branch, coordinate-wise minimization under $c \geq 0$ clips the negative entry: $c = e_0$ with $\|e_0 - t\|^2 = 1$. On the all-nonpositive branch, $(c_0 - 1)^2 \geq 1$ already gives $\|c - t\|^2 \geq 1$. Hence the floor is 1. $\square$

The non-negativity of S buys exact energy conservation but forbids subtraction (floor $\frac{1}{2}\,\mathrm{Var}(Y)$); the heterodyne readout keeps the energy budget unistochastic while releasing the sign of the mixing—probabilities are additive and non-negative, whereas amplitudes interfere. An inverse-free special case replaces U by $\frac{n(n-1)}{2}$ Givens rotations Q ($S = Q \odot Q$), a $O(n^2)$-parameter unistochastic subfamily (Sec. 3.1).

3 Empirical Results

All experiments: NVIDIA RTX 4060 Laptop GPU (8 GB), PyTorch 2.x, WSL Ubuntu 24.04, stream width $n = 4$; all baselines are our re-implementations (official code was not used).

3.1 Operator precision and speed

Table 1: Uni-mHC and the Givens variant satisfy the constraint to $\sim 10^{-7}$ (pure float32 rounding), whereas Sinkhorn–Knopp with $T = 20$ leaves a column-sum residual of $5.0e-03$—the price of an iterative projection. The Cayley solve is also the cheapest: one well-conditioned 4×4 complex solve instead of 20 normalization sweeps (Thm. 1iv).

3.2 Subtractive disentanglement

We generate fresh $X \in \mathbb{R}^{512\times4\times64}$, $\mathcal{N}(0, 1)$, per step and ask the mixing layer alone (shared learnable gain γ) to produce $\hat{y} = \gamma \sum_k W_{0k} X_k$ against $Y = X_0 - X_1$; 800 Adam steps. Figure 2: Uni-mHC drives the loss to $1.14e - 11$ and GivensMHC to $9.72e - 15$ with coefficients $(+1.00, -1.00, 0, 0)$—the exact destructive interference pattern of Theorem 2. Sinkhorn mHC and the orthostochastic baseline stall at $9.86e - 01$ and $9.85e - 01$ (floor: 1.0), their best solution being the trivial copy X_0. All gradients were finite throughout, confirming stable complex autograd.

3.3 Language modeling with four residual streams

We modify nanoGPT (Karpathy, 2023): each of the 6 blocks (`n_embd=384`, 6 heads, context 256) operates over $n = 4$ residual streams; every attention/MLP stage reads a learned combination

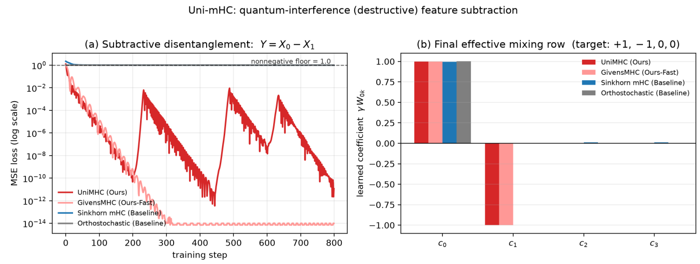

Figure 2: **Subtractive disentanglement** ($Y = X_0 - X_1$). **(a)** Uni-mHC and GivensMHC reach machine precision; non-negative baselines plateau at the impossibility floor (Prop. 2). **(b)** Learned coefficients γW_{0k}: phase interference yields the exact $(+1, -1, 0, 0)$ pattern; non-negative mixers degenerate to a pure copy $(+1, 0, 0, 0)$.

Table 2: **nanoGPT on** `shakespeare_char` (6 layers, 4 streams, 1000 steps). Val loss: mean of 50 eval batches; peak memory $< 2.3\,\text{GB}$.

Mixer	final train loss	final val loss	ms/iter
vanilla GPT (no HC)	0.112	3.459	2644
Uni-mHC (Ours)	1.248	1.513	384
GivensMHC (Ours-Fast)	1.251	1.523	383
Sinkhorn mHC	1.259	1.507	546
Orthostochastic	1.249	**1.505**	483

$\sum_k a_k x_k$, writes back with learned per-stream weights, and a manifold mixer redistributes the streams, $x^{(t+1)} = M^{(t)} x^{(t)} + b^{(t)} \otimes f_t(\sum_k a_k^{(t)} x_k^{(t)})$. Mixers start at identity, so training begins as a standard pre-norm GPT (batch 32, 1000 steps, AdamW, cosine schedule, `shakespeare_char`; all runs resumable). Table 2 reports final losses.

At this scale the vanilla GPT overfits within the short horizon (train 0.11, val 3.46), while every manifold-constrained variant—whose identity-initialized stream topology warms up effective capacity gradually—holds val ≈ 1.51. Stability is the second point: 12 mixers $\times$ 1000 steps with signed weights never destabilized training; post-training energy matrices violate double stochasticity by at most 7×10^{-7}, and all 12 mixers discovered signed interference weights—the unistochastic manifold is safely exploitable.

4 Conclusion

Uni-mHC replaces the projected doubly stochastic constraint of mHC with an exact unistochastic parameterization via a complex Cayley transform, keeping exact energy conservation and a non-expansive mixer while the heterodyne readout restores subtractive feature arithmetic through phase-π destructive interference—impossible for entrywise non-negative mixers, as we prove, and verified empirically (10^{-11} vs. 0.99). Future work: token-conditional phases, scaling studies, and links to quantum circuit compilation.

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